Bilal Riaz

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PhD graduate in Electrical and Computer Engineering with expertise in machine learning and applied optimization. I am seeking research roles where I can develop and deploy optimization and learning methods for signal processing and control, with an emphasis on impactful real-world applications. My research experience also includes generative modeling (GANs; flow- and diffusion-based image generation) and an active interest in large language models for scientific and industrial use cases.

Education

University of Delaware
PhD in Electrical and Computer Engineering *2019–2025*
Thesis: *Applications of Computational Optimal Transport in Machine Learning and Signal Processing*.
Advisor: Prof. Austin J. Brockmeier

University of Delaware
M.S. in Electrical and Computer Engineering *2019–2022*
Thesis: *On Spectral Clustering, Informativeness, and Seriation*.
Advisor: Prof. Austin J. Brockmeier

University of Science and Technology of China
M.S. in Control Engineering *2015–2018*
Thesis: *Preparation of Quantum Gates with Control-Theoretic Methods*.
Advisor: Prof. Cong Shuang

Pakistan Institute of Engineering and Applied Sciences
B.S. in Electrical Engineering *2010–2014*
Thesis: *Modeling and Control of a Self-Balancing Bicycle*.
Advisor: Prof. Muhammad Abid

Work and Research Experience

Department of Industrial Engineering, University of North Carolina at Charlotte
Postdoctoral Associate *2026–Present*
- Develop reinforcement learning–based methods to fine-tune pretrained diffusion models for antibody generation.
- Optimize diffusion sampling toward empirical design objectives, including improved antibody–target binding energy.
- Contribute to an industry-sponsored pharmaceutical project on computational antibody design.

Department of Electrical and Computer Engineering, University of Delaware
Research Assistant *2023–2025*
Developed optimal-transport methods for aligning continuous distributions for dataset alignment in an Office of Naval Research–funded project.

Department of Electrical and Computer Engineering, University of Delaware
PhD Student *2019–2025*
- Conducted research on computational optimal transport for machine learning and signal processing.
  1. Developed optimal-transport–based support subset selection algorithms and applied them to PU learning, semi-supervised learning, point-cloud registration, and color transfer (TMLR).
  2. Applied optimal transport to align collections of normalized signal spectra via monotone mappings (submitted to IEEE Open Journal of Signal Processing).
  3. Extended support subset selection to Monge maps for aligning datasets in neural-network latent spaces (manuscript in preparation).
- Studied the role of informativeness in seriation and spectral clustering.

Department of Electrical and Computer Engineering, University of Delaware
Research Assistant *Summer 2020*
Developed computational simulations of odor-plume dispersion in aquatic environments for an Office of Naval Research–funded project.

Huawei Pakistan
Project Controller *2018–2019*

Coordinated project planning and execution for 3G/4G network deployment and upgrades.

Department of Automation, University of Science and Technology of China
Master's Thesis Student *2015–2018*
Developed Lyapunov-based and optimal-control methods for computational quantum-gate design in the Systems Engineering Lab, leading to two conference papers and one journal publication.

Publications

[1] **Bilal Riaz**, Yuksel Karahan, and Austin J. Brockmeier. "Partial Optimal Transport for Support Subset Selection". In: *Transactions on Machine Learning Research* (2023).

[2] Yuksel Karahan, **Bilal Riaz**, and Austin J Brockmeier. "Kernel Landmarks: An Empirical Statistical Approach to Detect Covariate Shift". In: *NeurIPS 2021 Workshop on Distribution Shifts: Connecting Methods and Applications*. 2021.

[3] **Bilal Riaz**, Cong Shuang, and Shahid Qamar. "Optimal Control Methods for Quantum Gate Preparation: a comparative study". In: *Quantum Information Processing* 18 (2019), pp. 1–26.

[4] **Bilal Riaz**, Cong Shuang, and Shahid Qamar. "Lyapunov based Control for one qubit Quantum gates in Coherence vector formulation". In: *2018 37th Chinese Control Conference (CCC)*. IEEE. 2018, pp. 8116–8121.

[5] Shahid Qamar, Cong Shuang, and **Bilal Riaz**. "Lyapunov-based feedback control of two-level stochastic open quantum systems". In: *2017 IEEE International Conference on Cybernetics and Intelligent Systems (CIS) and IEEE Conference on Robotics, Automation and Mechatronics (RAM)*. IEEE. 2017, pp. 48–53.

[6] **Bilal Riaz** and Austin J. Brockmeier. "Neural Optimal Transport for Subset Alignment". (under review at ICLR 2027).

[7] Yuksel Karahan, **Bilal Riaz**, Hau Phan, Matthew S. Emigh, and Austin J. Brockmeier. "Finding Landmarks of Covariate Shift with Kernel Landmark Sliced Wasserstein Distance". (under review at AISTAT 2027).

[8] **Bilal Riaz** and Austin J. Brockmeier. "Optimal Transport with Frequency Warping for Bags of Spectra". Accepted at IEEE MLSP 2026.

[9] Yuksel Karahan, **Bilal Riaz**, Matthew S. Emigh, and Austin J. Brockmeier. "Fast Landmark Bures Distance for Interpreting the Divergence between Datasets". (under review).

Computational Skills

Numerical Computing: Proficient in Python for scientific computing (NumPy, SciPy, JAX); working knowledge of MATLAB.
Machine Learning Frameworks: Proficient in JAX, PyTorch, and scikit-learn; working knowledge of TensorFlow. Comfortable switching between frameworks and using automatic differentiation (PyTorch/JAX) to build large-scale, data-driven optimization pipelines.
Miscellaneous: Working knowledge of C++; proficient in LaTeX. Experience developing parallelizable machine-learning code for large-scale training and deployment. Comfortable with Bash and Slurm for HPC workflows.

Recommendations

Prof. Austin J. Brockmeier: PhD Advisor
Department of Electrical and Computer Engineering, University of Delaware

Prof. Xiang-Gen Xia: Dissertation Committee Member
Department of Electrical and Computer Engineering, University of Delaware

Prof. Dominique Guillot: Dissertation Committee Member
Department of Mathematical Sciences, University of Delaware